

Neural Particle Method: Experimental Record

Synthetic-data setup, machinery, and the benchmark suite

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Abstract

This note records, in one place, the synthetic markets and dynamics used throughout the project, the simulation, estimation and scoring machinery, and the specification of the clean benchmark suite agreed on 8 September 2026 (Section 4), run on 9 September 2026; its tables are generated from the MLflow store. The suite follows the discovery, in the diagnostics of 8 September, that all one-pass schemes were fitting the conditional expectation of the raw Euler variance rather than of its positive part, which the dynamics actually use (Section 3.4); the diagnostic runs themselves live in the MLflow experiments `mechanism`, `bump_floor` and `bump_floor_nn`.

Contents

1	Conventions	2
2	Synthetic data	2
2.1	Target surfaces: SSVI power law	2
2.2	Variance dynamics	2
2.3	Quotes, bump, and scoring	2
3	Machinery	3
3.1	Simulation	3
3.2	One-pass schemes and their estimators	4
3.3	Implicit loop	4
3.4	The variance-target discrepancy	4
4	Benchmark suite	4
4.1	Common setup	5
4.2	Cold suite	5
4.3	What a lag is	7
4.4	Offline bodies	7
4.5	Frozen body and online head	8
4.6	Reproducibility	9
5	Importance sampling	9
5.1	Slice layer	10
5.2	End to end	10
5.3	Far-wing quotes	12
5.4	The network and the density-weighted fit	12

1 Conventions

Explicit denotes the one-pass, per-slice calibration with a per-slice neural network. Nadaraya–Watson (NW) is the classical particle method: the same one-pass structure with kernel regression. Other per-slice estimators are named by their estimator (RKHS, spline, ridge). *Implicit* denotes the damped outer iteration with a single network in (t, x) . All errors are implied-volatility root-mean-square errors in volatility basis points against the target surface, unless stated otherwise. “Pooled” means over all quoted strikes and maturities together; “wings” means the quoted strikes with $|k| > 0.25$; per-maturity columns restrict to one maturity.

2 Synthetic data

2.1 Target surfaces: SSVI power law

Each scenario’s market is an SSVI implied-variance surface in the power-law parameterisation. With log-moneyness $k = \log(K/S_0)$, $S_0 = 1$, and parameters $(\sigma_0, \eta, \gamma, \rho_s)$,

$$\theta_T = \sigma_0^2 T, \quad \varphi(\theta_T) = \eta \theta_T^{-\gamma}, \quad w(k, T) = \frac{1}{2} \theta_T \left(1 + \rho_s \varphi k + \sqrt{(\varphi k + \rho_s)^2 + 1 - \rho_s^2} \right), \quad (1)$$

where w is total implied variance and the implied volatility is $\sqrt{w(k, T)/T}$. Parameters are drawn independently per scenario from

$$\sigma_0 \sim U(0.12, 0.35), \quad \eta \sim U(0.5, 2.0), \quad \gamma \sim U(0.3, 0.5), \quad \rho_s \sim U(-0.85, -0.10),$$

resampled until the sufficient no-arbitrage conditions $0 < \gamma \leq \frac{1}{2}$, $|\rho_s| < 1$, $\eta(1 + |\rho_s|) \leq 2$ hold. Scenario s_i uses the seed $1000 + i$ (`bench/scenarios.py`, `make_registry`).

The Dupire local volatility is analytic from Equation (1): with w_k, w_{kk}, w_T the derivatives of w ,

$$\sigma_{\text{Dup}}^2(T, k) = \frac{w_T}{1 - \frac{k}{w} w_k + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) w_k^2 + \frac{1}{2} w_{kk}}, \quad (2)$$

with the denominator floored at 0.05, the local variance clipped to $[10^{-8}, 9]$, and the time argument clipped to $[t_{\min}, T]$ with $t_{\min} = 0.004$ (`market/local_vol.py`). These clipping constants are part of the calibrated behaviour and are never changed.

2.2 Variance dynamics

The calibrated model is Heston-type LSV in log-spot $X = \log S$,

$$dX_t = -\frac{1}{2} L^2(t, X_t) V_t dt + L(t, X_t) \sqrt{V_t} dW_t, \quad dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dB_t, \quad d\langle W, B \rangle_t = \rho dt. \quad (3)$$

For scenario s_i the dynamics parameters are drawn from the same seed as the surface: $\kappa \in \{1, 2, 3\}$, $\xi \in \{0.3, 0.6, 1.0\}$, $\rho \in \{-0.7, -0.3\}$, each uniform over its set, and $\theta = v_0 = \sigma_0^2$ so that the variance level matches the surface’s short-end level. The Feller ratio $2\kappa\theta/\xi^2$ therefore ranges from 0.05 to 4.3 across the grid. Table 1 lists all twenty scenarios. Horizon $T = 2$ years; spot $S_0 = 1$; no rates or dividends.

Two further families exist in the registry and were not used today: six variants of s01 with (ξ, ρ) swept over the grid (`fig3_registry`), and three Heston-market scenarios whose target surface is generated by a Heston model with parameters different from the calibrated dynamics (`li_simple`, `li_complex`, `bayer`; horizon 1 year; Dupire surface from prices on a 50×121 grid).

2.3 Quotes, bump, and scoring

Quoted strikes are 13 log-strikes equally spaced in log between 0.6 and 1.6; maturities are 0.25, 0.5, 1 and 2 years. The wings are the strikes with $|k| > 0.25$ (seven of thirteen). The warm-start

sid	SSVI surface				dynamics			Feller
	σ_0	η	γ	ρ_s	κ	ξ	ρ	
s01	0.261	0.52	0.34	-0.21	1	0.3	-0.7	1.51
s02	0.208	1.04	0.45	-0.56	3	0.6	-0.7	0.72
s03	0.163	0.89	0.40	-0.35	3	1.0	-0.7	0.16
s04	0.120	0.70	0.33	-0.56	2	0.6	-0.3	0.16
s05	0.285	1.01	0.49	-0.72	2	1.0	-0.3	0.32
s06	0.304	1.00	0.38	-0.58	3	1.0	-0.3	0.55
s07	0.202	0.56	0.46	-0.49	3	1.0	-0.3	0.25
s08	0.158	1.72	0.34	-0.11	1	1.0	-0.7	0.05
s09	0.133	0.99	0.31	-0.64	3	1.0	-0.3	0.11
s10	0.142	1.30	0.48	-0.11	3	1.0	-0.7	0.12
s11	0.144	1.10	0.38	-0.11	2	1.0	-0.7	0.08
s12	0.191	1.01	0.33	-0.22	3	1.0	-0.7	0.22
s13	0.303	0.62	0.43	-0.51	2	1.0	-0.3	0.37
s14	0.220	1.38	0.48	-0.11	1	0.6	-0.3	0.27
s15	0.127	0.72	0.37	-0.33	3	0.3	-0.3	1.08
s16	0.172	1.16	0.37	-0.42	1	0.6	-0.7	0.17
s17	0.136	1.13	0.42	-0.50	1	0.3	-0.3	0.41
s18	0.257	1.18	0.47	-0.24	3	0.6	-0.3	1.10
s19	0.170	1.09	0.39	-0.23	2	0.3	-0.7	1.28
s20	0.253	0.93	0.32	-0.32	3	0.3	-0.7	4.27

Table 1: The twenty synthetic scenarios: SSVI surface parameters, variance dynamics, and the Feller ratio $2\kappa\theta/\xi^2$.

experiments bump the surface to $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, a level rise of one volatility point with curvature and skew nudged inside the no-arbitrage region.

Scoring reprices vanilla calls under the calibrated leverage field on fresh paths: 500 000 particles, 200 Euler steps over $[0, 2]$, seed 10 000 plus a strategy-specific offset, call price as the sample mean of the payoff, Black–Scholes inversion at the grid-snapped maturity, and the exact SSVI implied volatility at the same snapped time as the target (`pricing/reprice.py`, `pricing/metrics.py`). The bump experiments use 300 000 repricing paths. Errors are reported as RMSE in volatility bp, pooled, wings-only, and per maturity.

3 Machinery

3.1 Simulation

Calibration and repricing share one stepper. With $\Delta = T/K$, $v^+ = \max(v, 0)$, $z_1 = \rho z_b + \sqrt{1 - \rho^2} z_p$ and independent standard normals z_b, z_p ,

$$x_{k+1} = x_k - \frac{1}{2}L_k(x_k)^2 v_k^+ \Delta + L_k(x_k) \sqrt{v_k^+} \sqrt{\Delta} z_1, \quad v_{k+1} = v_k + \kappa(\theta - v_k^+) \Delta + \xi \sqrt{v_k^+} \sqrt{\Delta} z_b. \quad (4)$$

This is full truncation: the raw v may be negative and is carried, but only v^+ enters any coefficient. The paper calibrated with $K = 50$ steps; the benchmark suite of Section 4 uses 200 for calibration and repricing alike. The leverage is stored per slice on a grid in log-spot and evaluated by linear interpolation, constant beyond the grid ends; the slice in force at time t is the last grid time $\leq t$. The explicit and NW passes use a per-slice grid at the 0.1% to 99.9% quantiles of the current cloud (101 points); the implicit loop uses a fixed grid of 81 points from log 0.4 to log 2.2. The leverage is capped at $L_{\max} = 4$ and the fitted conditional expectation is floored at 10^{-4} .

3.2 One-pass schemes and their estimators

At slice k , fit $f_k(x) \approx \mathbb{E}[V_{t_k} \mid X_{t_k} = x]$ on a subsample of 30 000 particles from the current cloud (the earlier default of the code; the benchmark suite of Section 4 fits on the full cloud), set $L_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\text{max}})$, and advance the cloud. Slice 0 uses $f_0 \equiv v_0$. Estimators:

- **NW**: Gaussian Nadaraya–Watson with Silverman bandwidth $h = 1.06 \hat{\sigma}_x n^{-1/5}$ on the subsample.
- **Explicit (per-slice network)**: a $1 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network with softplus output scaled by 0.04, input scaled by 0.3, trained by Adam at learning rate 10^{-2} on the unweighted squared loss; 400 steps on the first fitted slice and 120 steps on each later slice, warm-started from the previous slice’s weights.
- **RKHS**: kernel ridge with a Gaussian kernel of variance 0.1 in log-spot, 100 centres at the $j \cdot 100/101$ percentiles of the cloud, regularisation $\lambda = 10^{-9}$ (Bayer et al.’s published choices).
- **Ridge head on a frozen body**: the implicit network’s body (below) as a 65-dimensional feature map, with a per-slice ridge readout, $\lambda = 10^{-3}$ per sample, shrunk toward the previous slice’s coefficients.

3.3 Implicit loop

A single network $f_\theta(t, x)$, $2 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU with softplus output scaled by 0.04, inputs $(t/T, x/0.3)$. Starting from the explicit output as L^0 , each outer iteration simulates a full cloud under L^n with fresh random numbers, draws a pooled subsample of 60 000 points across all slices, trains θ for 300 Adam steps at learning rate 10^{-2} with the network and optimiser state carried over from the previous iteration, reads $\Phi(L^n)_k = \text{clip}(\sigma_{\text{Dup}}(t_k, \cdot) / \sqrt{\max(f_\theta(t_k, \cdot), 10^{-4})}, 0, L_{\text{max}})$ on the fixed grid, and sets $L^{n+1} = (1 - \alpha)L^n + \alpha\Phi(L^n)$ with $\alpha = \frac{1}{2}$. The paper uses six iterations. The pooled data are truncated, v^+ , before training.

3.4 The variance-target discrepancy

Before 8 September 2026 the one-pass schemes handed the estimator the raw Euler v , while the implicit loop handed it v^+ . Since only v^+ drives the dynamics Equation (4), the quantity the leverage must divide by is $\mathbb{E}[V^+ \mid X]$. Under Feller violation a sizeable fraction of particles has $v < 0$ after a coarse step (on s11, $v_0 = 0.02$, $\xi = 1$, $\Delta = 0.04$: the one-step standard deviation of v is 0.028), so $\mathbb{E}[v \mid X] < \mathbb{E}[v^+ \mid X]$ by the mean negative excursion, the denominator was too small, and the leverage too large. The effect is independent of the particle count and shrinks with the step size. Fitting on v^+ (`ExplicitConfig.fit_v_floor`) has been the package default since 8 September 2026; every result in Section 4 and the study log (`paper/studies_log.md`) uses it, and the archived replays of the paper’s runs pin it off explicitly so that they still reproduce bit for bit.

4 Benchmark suite

This section specifies the clean benchmark agreed on 8 September 2026 and run on 9 September 2026 with the variance floor of Section 3.4 as the package default. Every cell of the tables below is one MLflow run, keyed by method, scenario, seed and, for the online stage, lag type and particle budget, in experiments `suite_cold`, `suite_offline`, `suite_lagged`, `suite_budget_tuned` and `pde_reference`. Every trained object is stored as a run artifact (network weights as a `torch` state dictionary, per-slice estimators and leverage fields as pickles and JSON) and can be reloaded by run id. The tables are generated from the store by `nparticle suite tables`.

4.1 Common setup

- **Scenarios.** The 20 SSVI scenarios of Table 1. The three Heston-market scenarios used in the literature comparisons (`li_simple`, `li_complex`, `bayer`; a Heston market surface, a different Heston variance process as the calibrated model, one-year maturity, log-strikes on $[-0.9, 0.9]$) are run in the same experiments and are in the store, but they score a single one-year maturity and are not reported in this draft.
- **Time grid.** 200 Euler steps for calibration and 200 for repricing, $\Delta = 0.01$ on $[0, 2]$, so that every method sits at its step-count plateau (the step-count diagnostics of 8 September 2026, MLflow experiment `mechanism`) and the comparison is about the estimator alone.
- **Particles.** 100 000 particles for every cold row and for the online stage at its full budget; at those sizes every per-slice fit and every online head sees the whole cloud, and the earlier code default of a 30 000-particle fit subsample is not used. The online budget is also swept below 100 000, and there too every fit uses the whole online cloud. Offline bodies at 200 000 and 500 000 particles: the fit subsample stays at 100 000, so each per-slice fit of a body reads a fresh 100 000-particle subsample of the offline cloud while the cloud itself is simulated at the full size.
- **Seeds.** Two seeds (0, 1) for every cold row and every online stage; one seed for each offline body. The reprice seed is 10 000 plus the calibration seed.
- **Floor.** All one-pass schemes fit $\mathbb{E}[V^+ | X]$; `fit_v_floor` becomes the default in this work and the tiny golden replays are regenerated in the same commit.
- **Implicit configuration.** Warm start from the explicit network field, 30 damped iterations at $\alpha = \frac{1}{2}$ with fresh noise per iteration, persistent network and optimiser (the best practical setting found in the convergence diagnostics of 8 September 2026, experiment `mechanism`).
- **Scoring.** As in Section 2.3: 500 000 repricing paths, Black–Scholes inversion at the snapped maturity, exact SSVI (or semi-analytic Heston) implied volatility as the target. Metrics per run: pooled and wings MAE and RMSE in volatility bp, both per maturity, the count of failed inversions, and the leverage RMSE against the PDE reference of the scenario. Errors are reported in implied-vol basis points; each table also shows the liquid-quote error (mean absolute error over $|k| \leq 0.25$ and $T \geq 0.5$) and the absolute option-price error in basis points of spot, recovered from the implied-vol errors through the Black–Scholes formula.
- **Latency.** Wall-clock seconds of the calibration only (repricing excluded). For the lagged suite, only the online stage is timed.
- **PDE reference.** The Fokker–Planck reference leverage (`reference/pde.py`) is solved per scenario at 200 steps and enters twice: repriced under the same Monte Carlo as every other row, which is the attainable floor of the scoring under this repricer (Gyöngy), and as the reference for the leverage-error column. Its repricing is repeated under a second reprice seed to measure the Monte Carlo noise floor of the score itself.

4.2 Cold suite

One calibration from scratch per method, scenario and seed at 100 000 particles. Methods, with their estimator of $\mathbb{E}[V^+ | X]$ per slice:

1. **NW:** Gaussian Nadaraya–Watson, Silverman bandwidth (Section 3.2).
2. **Explicit NN, short training:** per-slice network, warm-started across slices, 400 Adam steps on the first slice and 120 on each later slice, full batch (Section 3.2).
3. **Explicit NN, tuned:** the same per-slice scheme with a $1 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU network, minibatch Adam (batches of 8192, learning rate 10^{-3}), 2000 steps on the first slice and 500 warm-started steps on each later slice; the architecture and schedule that the study

log (paper/studies_log.md) found accurate in the wings and robust under Feller violation.

4. **Explicit NN, searched:** the per-slice scheme with the recipe found by the Optuna search recorded in the study log (paper/studies_log.md): a $1 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1$ SiLU network, minibatch Adam (batches of 8192, learning rate 3.7×10^{-3}), 500 steps on the first slice and 400 warm-started steps on each later slice, fitted on 250 000 particles of the cloud, with mean matching (the fitted slice is rescaled to reproduce the cloud’s mean variance), a monotone penalty on slopes of the wrong sign, and a linear continuation of the slice beyond its quantile grid.
5. **Implicit NN:** damped global-network loop (Section 3.3, configuration above).
6. **RKHS:** Gaussian kernel ridge with 100 quantile centres, Bayer et al.’s published knobs (Section 3.2).
7. **Spline:** penalised cubic B-spline, 25 quantile knots, $\lambda = 1$.
8. **Literature block.** GHL kernel: Nadaraya–Watson in spot with the quartic kernel and the Guyon–Henry–Labordère rule-of-thumb bandwidth $h(t) = 1.5 S_0 \sigma_{LV}(S_0, t) \sqrt{\max(t, 0.25)} N^{-1/5}$. Bins: piecewise-constant regression on 20 equal-frequency buckets (van der Stoep et al.). Muguruza: the kernel-free conditional Monte Carlo of Corollary 4.1, each particle contributing its one-step Gaussian conditional law of log-spot given the variance path. PU-RBF: Hakala’s partition-of-unity radial basis functions, 40 centres, 5-nearest-neighbour widths, regulariser 0.2.
9. **PDE:** the reference leverage repriced by Monte Carlo (attainable floor).

Table 2: Cold suite, 20 SSVI scenarios, mean and median over scenarios and two seeds. Liquid: $|k| \leq 0.25$, $T \geq 0.5$. The PDE row is the attainable floor of the scoring, its seed-to-seed spread is 6.6 bp (the Monte Carlo noise floor of the score), and its latency is the Fokker–Planck solve. Bold: the best method row in each column; the PDE floor is the reference and is not a contender.

method	implied vol, bp				price, bp	s
	mean	median	liquid	wings	pooled	median
NW	46	33	16	70	2.7	8.0
Explicit NN, short training	57	36	16	86	2.6	1042.7
Explicit NN, tuned	49	35	14	75	2.4	536.3
Explicit NN, searched	48	36	17	72	2.7	895.1
Implicit NN	58	37	19	88	3.3	1298.6
RKHS	81	70	45	113	8.5	12.4
Spline	47	34	14	73	2.5	2.7
GHL kernel	45	32	15	70	2.5	8.5
Bins	66	61	16	105	2.7	1.5
Muguruza	154	139	184	147	29.1	10.6
PU-RBF	57	40	16	90	2.8	2.5
PDE (attainable floor)	42	27	13	62	2.2	207.3

Results (Table 2). With the variance floor in place the classical kernels are the best cold calibrators: the GHL kernel and NW sit at 45–46 bp pooled, 3–4 bp above the attainable floor of 42, and the spline at 47. The tuned per-slice network closes most of the gap the short-trained network leaves (57 $\rightarrow$ 49 bp pooled, 86 $\rightarrow$ 75 in the wings) and is the best row on liquid quotes (14 bp, against 15–16 for the kernels and 13 for the floor) and in price space (2.4 bp of spot against 2.5–2.7), but it does not beat the kernels on the pooled score: it is ahead of NW on 12 of the 20 scenarios, by a couple of basis points each, and behind by 14–39 bp on s03, s08 and s11, the scenarios where the variance spends most of its time at zero. Its latency is 536 s against 8 for NW, so as a cold calibrator it is a match, not an improvement. The searched network (study log) is 48 bp pooled, between NW and the hand-tuned network, better in the wings (72) and worse on liquid quotes (17), at 895 s: the search improved the estimator of $\mathbb{E}[V | X]$ by a few percent, and that does not change the cold picture. The implicit loop is 58 bp and the slowest row. Bins, PU-RBF, RKHS and Muguruza’s

conditional Monte Carlo are behind the kernels by 10 to 110 bp, RKHS and Muguruza mostly in the wings. Every row is dominated by the three-month wings; on liquid quotes all the competitive rows are within 3 bp of the floor, and in price space within 0.5 bp of spot.

4.3 What a lag is

An overnight calibration is done on surface S_0 ; the intraday market is S_1 , and the question is how much of the overnight work survives. Because the synthetic surfaces are parametric, a lag is a move in parameter space and never a bump of quoted points, so arbitrage cannot be introduced by accident. Two lags are run for every cell.

Surface lag. For SSVI, the existing bump: $\sigma'_0 = \sigma_0 + 0.01$, $\eta' = 0.95\eta$, $\rho'_s = \min(\rho_s + 0.03, -0.05)$, shrunk by quarters until the Gatheral–Jacquier sufficient conditions ($\eta(1 + |\rho_s|) \leq 2$, $0 < \gamma \leq \frac{1}{2}$) hold; calendar no-arbitrage is automatic since $\theta_T = \sigma_0^2 T$ is increasing. For a Heston market, the matching move in vol terms: $\sqrt{v_0}$ and $\sqrt{\theta}$ up one volatility point, $\rho \rightarrow \min(\rho + 0.03, -0.05)$, $\xi \rightarrow 0.95\xi$; the surface is model-generated and arbitrage-free for any valid parameters.

Surface plus spot lag. The surface lag together with a sticky-strike spot move of +2%, $S_1 = S_0 e^\delta$, $\delta = \log 1.02$. Under sticky-strike the implied volatility at each absolute strike is unchanged, so in log-moneyness relative to the new spot the surface is translated: $\sigma'_{IV}(k, T) = \sigma_{IV}(k + \delta, T)$. A translation in k preserves Durrleman’s butterfly condition (it involves only w and its k -derivatives at a point) and calendar monotonicity, so the shifted surface is arbitrage-free. Call prices at fixed strike change with the spot, so the Dupire local volatility is that of the translated surface, $\tilde{w}(k) = w(k + \delta)$ with k measured from the new forward; the particles also start at the new spot. The stored denominators $f(t, x) = \mathbb{E}[V_t | X_t = x]$ were fitted for a cloud started at S_0 , and a conditional expectation given spot depends on the starting point: with $\rho < 0$ a particle sitting 2% above the old start had its variance drift down, so the stale f at the new spot is below v_0 , the stale leverage is too high near the money, most at short maturities. Sticky-delta, by contrast, leaves the surface in moneyness unchanged and the whole calibration is scale invariant, so a sticky-delta spot move is invisible to every method; the surface-only lag already covers it. A 2% daily move is one to one-and-a-half standard deviations at 20–30% volatility.

4.4 Offline bodies

Offline stage (experiment `suite_offline`, one run per scenario, body type and particle count): a full calibration on S_0 at 200 000 or 500 000 particles, 200 steps, seed 0. Three bodies are trained. The **explicit NN, short training** is the per-slice network pass on the paper’s schedule (400 Adam steps on the first slice, 120 on each later slice, full batch); the **explicit NN, tuned** is the same pass with the network and schedule of item 3 of Section 4.2 ($1 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$ SiLU, minibatch Adam, 2000 steps on the first slice and 500 on each later slice); both save the per-slice networks and the per-slice denominators f_k . The **implicit NN** is the damped global loop of Section 3.3 and saves the global network. Each body is scored here as a cold calibrator on its own overnight surface S_0 , against the same repricer and the same quotes as Section 4.2, so the table says what a body is worth before any lag. These runs are cached and reused by every online cell below.

Results (Table 3). The tuned network is the only body worth keeping: 47 bp pooled at either size, against 54–55 for the short-trained network and 59–62 for the implicit loop, and it trains in half their time. More offline particles do not help any body at these training lengths (200k and 500k are within 1 bp), and the implicit loop gets slightly worse with more particles; the fit, not the cloud, is the limit. The tuned body’s liquid-quote error (14 bp) is within 1 bp of the floor. The searched body at 500 000 particles is 48 bp pooled, the best of the bodies in the wings (70) and the worst on liquid quotes (18); mean matching moves the near-money level while it fixes the wings. It trains in 856 s. (Only the 500 000-particle searched body was trained; it is the one the online head uses.)

Table 3: Offline bodies as cold calibrators on S_0 , 20 SSVI scenarios, seed 0; training time is the full offline calibration. Bold: the best method row in each column; the PDE floor is the reference and is not a contender.

body	particles	pooled	liquid	wings	price, bp	s
Explicit NN, short training	200k	54	15	83	2.6	1047.2
Explicit NN, short training	500k	55	16	84	2.7	1049.0
Explicit NN, tuned	200k	47	14	72	2.5	529.1
Explicit NN, tuned	500k	47	14	73	2.4	535.2
Explicit NN, searched	200k	—	—	—	—	—
Explicit NN, searched	500k	48	18	70	3.0	855.7
Implicit NN	200k	59	17	93	2.9	1365.5
Implicit NN	500k	62	20	95	3.3	1469.2
PDE (attainable floor)	—	42	13	62	2.2	207.3

4.5 Frozen body and online head

Online stage (one run per scenario, method, lag, online budget and online seed): given the intraday surface S_1 and a fresh cloud, build the intraday leverage and reprice. Only this stage is timed. The body is the tuned 500 000-particle body of Section 4.4, frozen; the methods differ in what is refitted online:

1. **Tuned body, stale f , fresh Dupire:** $L_k = \text{clip}(\sigma_{\text{Dup}}^{S_1}(t_k, \cdot) / \sqrt{\max(f_k, 10^{-4})}, 0, L_{\max})$ with the offline per-slice f_k . Costs nothing online beyond the repricing.
2. **Tuned body with spline head:** simulate the online particles under the stale- f leverage of item 1 with fresh Dupire, slice by slice; at each slice fit the penalised cubic B-spline of Section 3.2 (25 quantile knots, $\lambda = 1$) to the residual $v^+ - f_k(x)$ on the online cloud, and use $\max(f_k + \text{residual}, \frac{1}{4}f_k)$ as the denominator for the next step; the floor keeps the head from driving the leverage to its cap on Feller-violating scenarios.
3. **Tuned body with RKHS head:** the same residual head with a Gaussian kernel ridge (kernel width 0.5 times the cloud’s standard deviation, ridge 10^{-3}) in place of the spline, with the same floor.
4. **Tuned body with ridge head:** the per-slice network body frozen as a feature map (64 features plus a constant) with its last layer refitted by ridge on the online cloud, shrunk toward the offline readout. The body’s output is $\text{softplus}(Aw) \cdot 0.04$, and the refit minimises $\sum_i (\text{softplus}(A_i w) \cdot 0.04 - v_i^+)^2 + \lambda_{\text{eff}} \|w - w_0\|^2$ on the raw variance scale by Gauss–Newton to convergence (at most 20 steps) from the offline last layer w_0 , with $\lambda_{\text{eff}} = \lambda$ times the mean diagonal of the Gauss–Newton curvature at w_0 and $\lambda = 10^{-3}$, so the penalty is a fixed fraction of the data curvature whatever the output scale and cloud size. The denominator is $\text{softplus}(Aw) \cdot 0.04$ at the converged w . When the online data agree with the offline fit the first step is zero and $w = w_0$.

The online reference row is NW re-solved from scratch on S_1 at the same budget and 200 steps (the best intraday strategy in the floor-corrected bump runs, experiment `bump_floor`). The online budget is varied over 10 000, 30 000, 80 000 and 100 000 particles, with every fit on the whole online cloud: the 100 000 column is experiment `suite_lagged`, the other three are `suite_budget_tuned` (`scripts/budget_sweep.py`). The same four heads on the short-trained explicit body and on the implicit body are also in the store (experiment `suite_lagged`) and are summarised in the study log (`paper/studies_log.md`).

Results (Table 4). A frozen tuned body with a guarded online head is at least as good as a fresh NW calibration at every online budget, and better where the budget is small. At 10 000 online particles NW re-solved from scratch is at 50 bp under both lags; the spline head is at 46, the RKHS and ridge heads at 46, and the advantage shrinks as the online cloud grows: at 80 000 particles, the practical budget, the spline head is 43 against NW’s 44–45 under both lags, at the same online time (7 s),

Table 4: Frozen tuned or searched body (500 000 particles) with an online head, against NW re-solved from scratch, by online particle budget: pooled MAE in vol bp, 20 SSVI scenarios, two online seeds, both lags; liquid MAE at 100 000; online seconds at 10 000 and 100 000 particles (median). Bold: the best method row in each column; the PDE floor is the reference and is not a contender, and a zero online time (no online work) does not count.

method	surface lag					surface plus 2% spot lag					online s	
	10k	30k	80k	100k	liquid	10k	30k	80k	100k	liquid	10k	100k
NW re-solve on S_1	50	47	44	44	17	50	47	45	44	18	1.0	8.5
Tuned body, stale f , fresh Dupire	50	50	50	50	28	88	88	88	88	56	0.0	0.0
Tuned body + spline head	46	45	43	43	16	46	46	44	43	16	1.2	9.8
Searched body + spline head	46	45	42	42	16	46	45	43	43	16	1.0	7.1
Tuned body + RKHS head	46	46	44	44	16	46	46	44	44	16	2.3	18.8
Tuned body + ridge head	46	45	43	44	15	46	45	43	43	16	3.3	28.4

and the ridge head is the best row after the spot move (43) at three times the time; at 100 000 every head is within 1–2 bp of NW and 1 bp of each other. On liquid quotes the heads are at 15–16 bp against NW’s 17–18 at every budget. The head is doing the work: the stale body with a fresh Dupire numerator is 50 bp under the surface lag and 88 after the spot move, at every budget, because the stored $\mathbb{E}[V | X]$ was fitted for a cloud started at the old spot. The spline head is the row to use: the cheapest head, never worse than the others, and it turns the tuned body’s cold-suite draw into a small but consistent win over the re-solve at every budget from 10k to 100k. The searched body with the spline head is the best row at every budget from 30 000 up: 42 bp under the surface lag at 80 000 and 100 000 against NW’s 44, and 43 after the spot move against 45 and 44, with the liquid-quote error at 16 bp against NW’s 17–18, and the fastest online sweep (7.1 s at 100 000 against 8.5 for NW), because the network is half the width. Its liquid-quote deficit as a cold calibrator disappears once the head runs on it. At 10 000 the searched and tuned bodies tie (46 bp). The search bought about half a basis point over the hand-tuned body at the practical budget and a faster head; the two-basis-point margin over the re-solve is the body-plus-head design, not the search.

4.6 Reproducibility

Each run records the git hash, the full configuration as parameters, the leverage field, the implied-volatility error grid, and the trained objects. A loader (`nparticle suite load <run_id>`) rebuilds the leverage field and, where present, the network or per-slice estimators from the artifacts alone, so any row of the tables can be re-scored or reused as an offline body without recomputation.

5 Importance sampling

The particle method estimates $\mathbb{E}[V_t | X_t]$ from wherever the particles happen to be, and at short maturities almost none of them are in the wings. A volatility-preserving Girsanov tilt moves them there: the cloud is simulated as a defensive mixture of three drifts $-\theta, 0, +\theta$ on log-spot with weights $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, the exact discrete-time weights are accumulated along each path and bounded by 2, and every per-slice fit becomes a weighted regression (`calibrate/importance.py`). The online cost is one drift term and three running scalars per particle. This section measures what the tilt buys, where, and what stops it buying more.

The summary is this. For the classical kernel estimator the tilt removes a fifth to a third of the wing variance at every particle budget and lowers the leverage error against the PDE reference by 12 to 14 percent, so that 30 000 tilted particles match 80 000 untilted ones on both. On the quoted surface that is worth 3 bp at 80k. On far-wing quotes, where the tilt changes the leverage, it is worth 4 bp for the kernel method and 2.4 bp for the frozen body with an online head, at no online cost. The network estimator does not benefit until its loss is reweighted by the sampling density, after which it does.

A design is a schedule for θ and a cap c on $\sum_k \theta_k^2 \Delta t / (1 - \rho^2)$, the log-weight variance of a pure tilt. A constant schedule has $\theta = \sqrt{c(1 - \rho^2)/T}$; an inverse-square-root schedule, $\theta_k = a/\sqrt{\max(t_k, \Delta t)}$ with a fixed by the same cost, concentrates the tilt at short times; a front-loaded schedule spends the whole cost on $[0, 0.25)$ and then stops. The sweep is $c \in \{1, 3, 9\}$ for the first two schedules and $c \in \{3, 8\}$ for the third, plus untilted.

5.1 Slice layer

The tilt acts on estimation variance, so it is measured first on its own, without the feedback of the calibration loop. Particles are simulated under the frozen PDE-reference leverage of each of the six study scenarios (s01, s02, s05, s09, s11, s16), and at the four quoted maturities NW and the searched network are fitted on the weighted cloud and evaluated at the 13 quoted strikes against the PDE conditional expectation, at 10k, 30k and 80k particles with five seeds. The error at each strike splits into its standard deviation over seeds, which the tilt can remove, and its mean, the shared Euler-versus-Fokker-Planck bias, which it cannot. NW’s bandwidth is Silverman’s rule on the weighted spread of the cloud, so both arms smooth on the same scale. A design wins on the smallest NW wing standard deviation at 30k, and is disqualified if the bias it adds exceeds one percentage point plus half the standard deviation it removes.

Table 5: Slice layer: wing standard deviation over seeds (at $T = 0.25$ and pooled over maturities) and wing bias, in percent of $\mathbb{E}[V | X]$, by estimator, budget and design (experiment `tilt_slices`). Bold: smallest in each standard-deviation column; the bias columns are signed and are not bolded.

estimator, N	none			constant-1			constant-3			constant-9			inverse-sqrt-1			inverse-sqrt-3			inverse-sqrt-9			front-3			front-8		
	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias	std $T=0.25$	std	bias
NW, 10k	25.5	13.6	-8.2	26.9	12.5	-8.3	23.6	11.7	-7.9	18.7	10.7	-7.4	22.1	11.5	-8.2	21.7	12.0	-7.7	14.9	11.0	-7.6	14.7	10.7	-7.4	14.8	11.2	-7.0
NW, 30k	19.3	10.4	-6.2	18.4	10.0	-6.5	19.6	9.3	-6.5	14.4	8.0	-6.3	19.3	9.3	-6.9	15.7	8.9	-6.3	14.4	8.5	-6.6	14.4	8.5	-6.4	11.2	8.6	-6.1
NW, 80k	20.0	8.8	-4.6	15.1	7.7	-6.7	11.7	6.5	-7.1	11.6	6.2	-6.1	10.5	6.6	-7.1	13.0	6.8	-6.2	11.9	6.9	-6.0	9.0	6.4	-5.8	7.7	6.3	-5.8
Net, 10k	11.6	8.1	-9.8	7.8	7.4	-9.4	7.5	7.4	-9.7	7.5	7.5	-10.5	7.9	7.1	-9.3	7.6	7.4	-10.0	7.3	7.7	-11.2	7.3	7.2	-9.7	7.9	8.1	-10.9
Net, 30k	5.4	5.1	-9.5	5.1	6.0	-9.5	5.3	5.7	-9.9	4.1	5.5	-10.1	5.0	5.7	-10.0	4.8	6.0	-10.7	4.0	5.7	-12.0	3.5	5.4	-11.1	3.9	5.8	-12.2
Net, 80k	8.2	6.3	-8.9	6.7	5.3	-9.8	6.0	5.2	-10.2	6.1	5.3	-11.0	6.4	5.2	-10.0	6.3	5.3	-10.5	5.9	5.5	-11.7	5.6	4.8	-10.7	3.3	4.6	-11.8

The winner is the strongest constant tilt, $c = 9$ (Table 5, Figures 1 and 2). For NW it takes the pooled wing standard deviation from 13.6, 10.4 and 8.8 percent at 10k, 30k and 80k to 10.7, 8.0 and 6.2, and the three-month figure from 25.5, 19.3 and 20.0 to 18.7, 14.4 and 11.6. Tilted 30k (8.0) sits at untilted 80k (8.8), which is the budget equivalence the tilt was built for. The bias is untouched, -6 to -8 percent at every design. The front-loaded schedule reaches the three-month wings better still (NW’s three-month standard deviation falls to 14.8, 11.2 and 7.7, and the far right strike goes from 13 effective particles to 102) at the same pooled figure, but its frozen sample fraction of 0.6 costs every later slice end to end, so the constant design is the one carried forward.

5.2 End to end

Cold: NW on the 20 SSVI scenarios and the searched network on the six study scenarios, at 10k, 30k and 80k particles, seeds 0 and 1, untilted and under the winning design, scored as Section 4.2 (experiment `tilt_cold`). Online: the searched 500 000-particle body with the spline head on a tilted online cloud at 10k and 80k, both lags (experiment `tilt_online`), against the untilted cells of Table 4; the P-spline is fitted by weighted least squares. Repricing is untilted throughout.

For NW the tilt lowers the leverage RMSE by 12 to 14 percent at every budget (0.124, 0.102, 0.092 $\rightarrow$ 0.108, 0.089, 0.082 at 10k, 30k, 80k), better in 39 of 40 scenario-seed pairs at 80k, and tilted 30k again matches untilted 80k. On the quoted surface it is worth 3 bp at 80k (46 $\rightarrow$ 43 bp pooled, 69 $\rightarrow$ 65 in the wings, better in 32 of 40 pairs), is within noise at 10k, and is 2 bp worse at 30k, where the loss sits at three months (98 $\rightarrow$ 107 bp, $t = 3.6$) while the leverage error still improves; the mechanism behind that last point is not settled. The frozen body with the spline head is unchanged at every budget and lag (42–43 bp at 80k either way): that row already sits on the floor of the quoted surface. The plain searched network gains nothing; Section 5.4 says why and fixes it.

Two things cap what the tilt can do on the quoted surface, and both are measurable from the stored runs. The three-month right side is bias, not variance: at $k = 0.14$ to 0.31 the slice-layer error in

Table 6: Cold calibration with and without the tilt by budget: pooled, wings, three-month and liquid MAE in vol bp, leverage RMSE against the PDE reference, the minimum per-slice effective sample fraction, and median seconds. NW rows are over the 20 SSVI scenarios; the network rows are over the six study scenarios, as their labels say, and the floor row is over the 20. Bold: smallest in each column over all rows but the floor, so a bold tilted row at a small budget reads as budget equivalence.

method, N , arm	pooled	wings	$T=0.25$	liquid	lev. RMSE	min ESS
NW (20 scenarios), 10k, untilted	51	74	107	21	0.124	–
NW (20 scenarios), 10k, tilted	50	73	107	20	0.108	0.58
NW (20 scenarios), 30k, untilted	45	66	98	18	0.102	–
NW (20 scenarios), 30k, tilted	47	70	107	17	0.089	0.58
NW (20 scenarios), 80k, untilted	46	69	105	16	0.092	–
NW (20 scenarios), 80k, tilted	43	65	101	16	0.082	0.58
Explicit NN, searched (six study scenarios), 10k, untilted	58	83	112	22	0.252	–
Explicit NN, searched (six study scenarios), 10k, tilted	60	85	115	23	0.216	0.58
Explicit NN, searched (six study scenarios), 30k, untilted	53	76	106	21	0.188	–
Explicit NN, searched (six study scenarios), 30k, tilted	59	85	114	20	0.204	0.58
Explicit NN, searched (six study scenarios), 80k, untilted	54	76	104	20	0.167	–
Explicit NN, searched (six study scenarios), 80k, tilted	54	78	109	19	0.180	0.58
Explicit NN, density-weighted (six study scenarios), 10k, untilted	57	79	112	25	0.267	–
Explicit NN, density-weighted (six study scenarios), 10k, tilted	51	70	106	22	0.219	0.58
Explicit NN, density-weighted (six study scenarios), 30k, untilted	48	64	95	24	0.207	–
Explicit NN, density-weighted (six study scenarios), 30k, tilted	54	73	108	24	0.184	0.58
Explicit NN, density-weighted (six study scenarios), 80k, untilted	50	68	103	21	0.157	–
Explicit NN, density-weighted (six study scenarios), 80k, tilted	49	66	98	23	0.168	0.58
PDE (attainable floor)	42	62	110	13	0.000	–

Table 7: Searched body with the spline head on an untilted and a tilted online cloud, by online budget and lag: pooled, wings and liquid MAE in vol bp and online seconds.

budget, lag, arm	pooled	wings	liquid	online s
10k, surface, untilted	46	67	19	1.0
10k, surface, tilted	45	65	19	1.1
10k, surface_spot, untilted	46	67	19	1.0
10k, surface_spot, tilted	47	69	20	1.1
80k, surface, untilted	42	63	16	6.1
80k, surface, tilted	42	62	16	6.7
80k, surface_spot, untilted	43	64	16	6.1
80k, surface_spot, tilted	43	64	16	6.6

$\mathbb{E}[V \mid X]$ is -16 to -19 percent for every design, with a standard deviation of 1 to 6, the Euler cloud disagreeing with the reference where V sits at its zero atom. And the scoring floor at 200 steps is mostly time-discretisation bias. Over the 20 scenarios the PDE-exact leverage repriced at 400 steps scores 28.4bp pooled, 43.2 in the wings, 63.8 at three months and 8.5 on liquid quotes, against 42.0, 61.9, 110.3 and 13.4 at 200 steps; NW itself, calibrated and repriced at 400 steps, goes from 45.6 to 34.3bp pooled at 80k (experiments `suite_pde_floor_s400`, `tilt_cold_s400`). On the six hardest scenarios NW at 200 steps even sits below the “floor” (52 against 55), because a method calibrated on the discrete scheme compensates part of the scheme’s bias while the PDE-exact leverage cannot; at 400 steps the floor is a floor everywhere. The tilt survives the finer step: at 400 steps it still lowers NW’s leverage error in 39 of 40 pairs and the pooled score by 1.4bp ($t = -3.3$). The time step, not the estimator, is the lever on the absolute numbers of Section 4, and every stored body interpolates in time, so section 4 can be re-scored at 400 steps without retraining.

5.3 Far-wing quotes

The quoted grid stops at $|k| \approx 0.5$ and its three-month wings fail inversion, so the score of Section 5.2 never sees the region where the tilt changes the leverage. Here every stored field is re-scored, without recalibration, on 21 log-strikes from 0.45 to 2.2 spot ($|k|$ up to 0.8) with an importance-sampled out-of-the-money repricer: puts below the money, calls above, on a cloud tilted by the constant design of cost 3 with exact discrete-time weights, inverted through put-call parity. A quote whose target price is below 10^{-5} of spot is not scored, which removes the three-month far wings. The far-wing error is the mean absolute implied-vol error over the scored quotes with $|k| > 0.25$.

Table 8: Far-wing implied-vol MAE in bp at the 80k budget: pooled over maturities and at 0.5, 1 and 2 years, the mean number of scored far-wing quotes, and the 13-strike pooled MAE for reference. Every row is over the same 20 SSVI scenarios and two seeds. Bold: smallest in each column over the non-floor rows.

row	far wings	$T=0.5$	$T=1$	$T=2$	quotes	pooled (13-strike)
NW, 200 steps, untilted	37	46	35	19	40	46
NW, 200 steps, tilted	32	40	31	18	40	43
NW, 400 steps, untilted	25	31	25	15	40	34
NW, 400 steps, tilted	24	29	23	16	40	33
Explicit NN, searched, untilted	38	48	35	20	40	48
Explicit NN, searched, tilted	38	47	36	20	40	49
Explicit NN, density-weighted, untilted	34	47	34	18	40	45
Explicit NN, density-weighted, tilted	33	44	30	18	40	44
NW re-solve (online)	37	47	35	20	42	44
Searched body + spline head, untilted	34	44	29	18	42	42
Searched body + spline head, tilted	32	43	26	15	42	42
PDE floor, 200 steps	21	29	15	8	40	42
PDE floor, 400 steps	13	18	10	7	40	28

This is where the tilt pays (Table 8). For NW at 80k the far-wing error falls from 36.6 to 32.5 bp, -4.1 bp paired over 40 scenario-seed pairs ($t = -5.9$, better in 34), at every maturity ($45.7 \rightarrow 39.6$ at six months, $35.4 \rightarrow 31.0$ at one year, $18.9 \rightarrow 17.6$ at two), a quarter of its gap to the 200-step floor of 20.7. The frozen searched body with the spline head on a tilted online cloud improves by 2.4 bp under both lags ($t = -3.9$ and -3.6 , better in 30 of 40), and that row beats the NW re-solve at the same budget by 5.5 to 5.9 bp ($t = -6.9$ and -8.1): the offline body carries the wings and the tilted online cloud corrects them. At 400 steps NW’s far-wing error is 25.2 bp untilted, 11 bp from the time step alone, and the tilt is still worth 1.5 bp on top ($t = -2.2$, better in 28 of 40). The plain searched network sits at NW’s untilted level (37.7) and does not move.

5.4 The network and the density-weighted fit

On s11 at three months the searched network’s $\mathbb{E}[V | X]$ is exactly zero from $k = 0.14$ outward, so its leverage sits on the cap $L_{\max} = 4$ across the right wing (Figure 2), while NW on the same cloud gives 0.007 to 0.056 against the PDE’s 0.012 to 0.064. No prior is responsible; switching the monotone penalty, the mean matching and the tail rule off changes nothing. The cause is the loss. About 900 of 80 000 particles sit at $k > 0.2$, and least squares gives that wing an influence equal to its probability mass, one percent, while the bulk, where f falls steeply to zero near $k = 0.06$, sets the shape; the network extrapolates the descent and its softplus output clamps at zero. NW is local and never sees the bulk from the wing. The same fact is why the tilt cannot help the network as built: it puts 1475 particles in that region instead of 906, each with an importance weight below one, and the weighted loss leaves the wing’s total weight at its physical mass, whereas NW’s local ratio gains from the particle count directly.

The fix is to weight the loss by $w(x)/\hat{p}(x)$, the importance weight divided by a kernel estimate of the physical density of X at the particle, floored at $1/50$ of the modal density and normalised to mean one (`NNRegressor(design_weight=True)`, cold row `explicit_nn_opt_dw`, the promoted

recipe otherwise unchanged). Any positive weight that depends on x alone leaves the minimiser at $\mathbb{E}[V \mid X]$, so the fit stays unbiased and the wings count by their particles. On the s11 slice the network then recovers the whole right wing (0.008, 0.020, 0.032, 0.040, 0.043 at the five outer strikes), smoother than NW. In the slice layer at 80k its three-month wing bias halves (-14.6 to -7.0 percent) with a third of NW’s three-month variance (6.3 against 20.0), at the cost of pooled wing variance (6.3 to 7.2), and the tilt now reaches it (three-month standard deviation $6.3 \rightarrow 4.7$).

End to end at 80k on the 20 scenarios (Tables 6 and 8, 40 pairs), the density-weighted network is 3.6 bp better than the plain one pooled ($t = -2.9$), 7.9 in the wings ($t = -3.1$) and 3.3 in the far wings ($t = -2.2$). Against NW it is a draw pooled (44.6 against 45.6), 5.6 bp better in the wings (63.4 against 69.0, $t = -1.9$), 2.2 better in the far wings ($t = -2.0$), and 3.8 bp worse on liquid quotes (19.7 against 15.9, $t = 3.4$). Equalising the loss across log-spot buys the wings with near-the-money accuracy, and the cap on the density weight, set at one value here, is the dial between the two. Under the tilt the density-weighted network is the best cold row in the wings (60.4 bp against NW’s 65.0), ties NW pooled (43.6 against 43.4) and in the far wings (32.5 each), and the tilt now moves it in the far wings by -1.9 bp ($t = -2.7$). At 30k on the six study scenarios it is again the best untilted cold row (48.1 bp against NW’s 52.0 and the plain network’s 53.3). What remains of its deficit to NW is in the bulk (leverage RMSE 0.187 against 0.149), and the offline body of Section 4.5 has not yet been retrained with this loss.

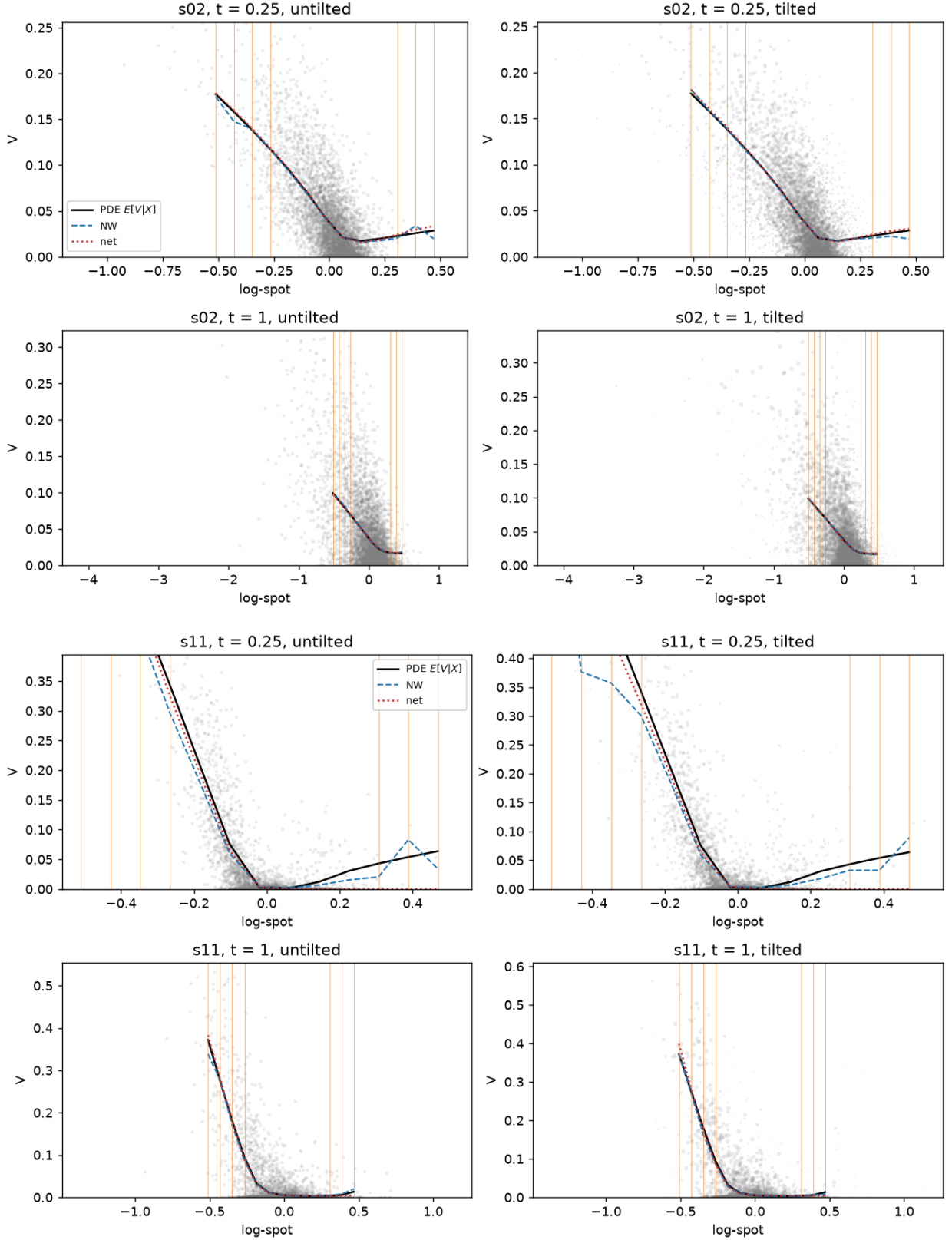


Figure 1: Particle clouds at 30k under the frozen PDE leverage, untitled (left) and under the winning design (right), at $t = 0.25$ (top) and $t = 1$ (bottom), with the PDE conditional expectation (solid), the NW (dashed) and network (dotted) estimates, and the quoted wing strikes; marker area is the importance weight. s02 (Feller satisfied) and s11 (Feller violated).

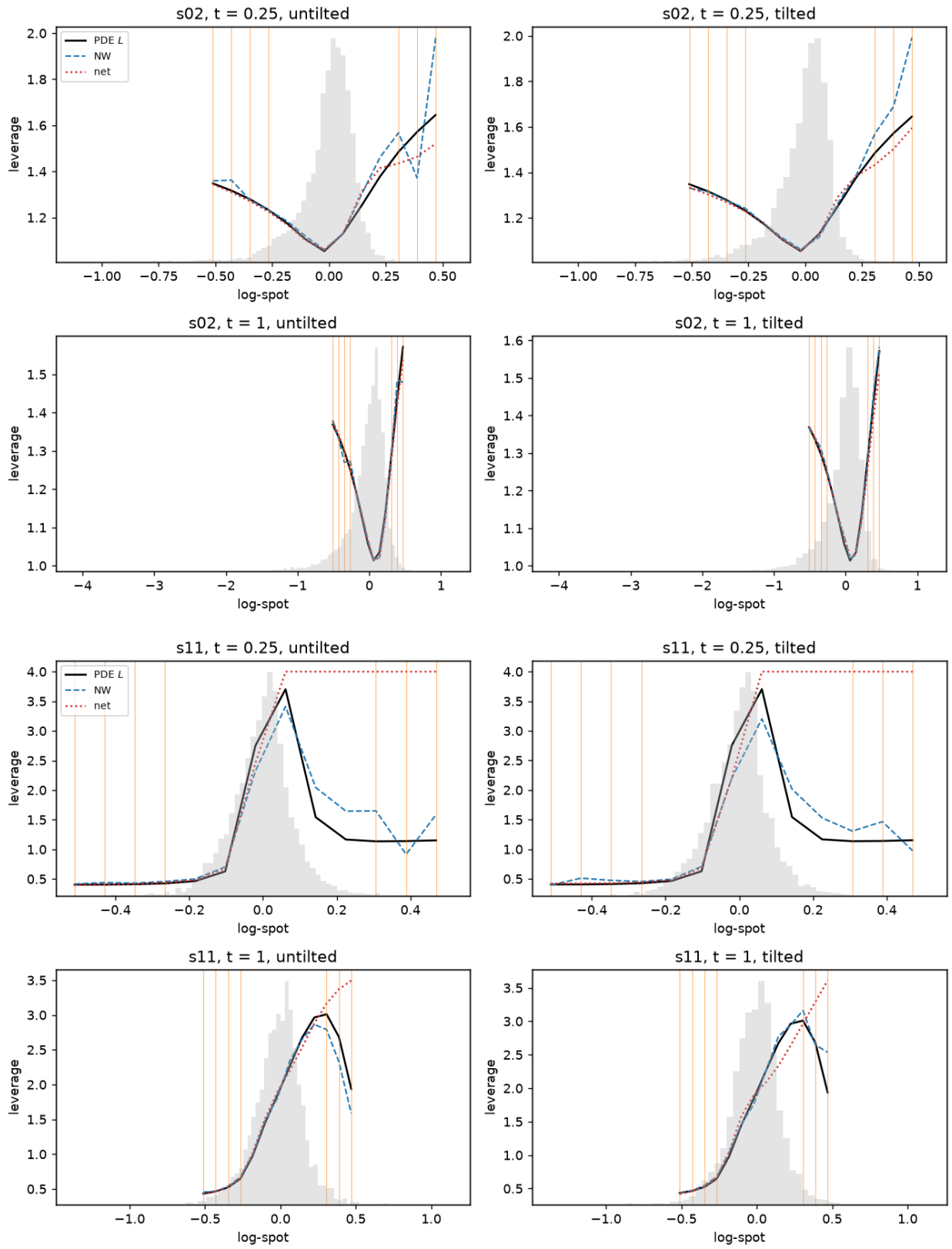


Figure 2: The same slices in leverage: PDE (solid) against the NW and network estimates, with the weighted marginal density of the cloud behind.